



DFSMS Advanced Topics: Tape Policy management

Session 2419

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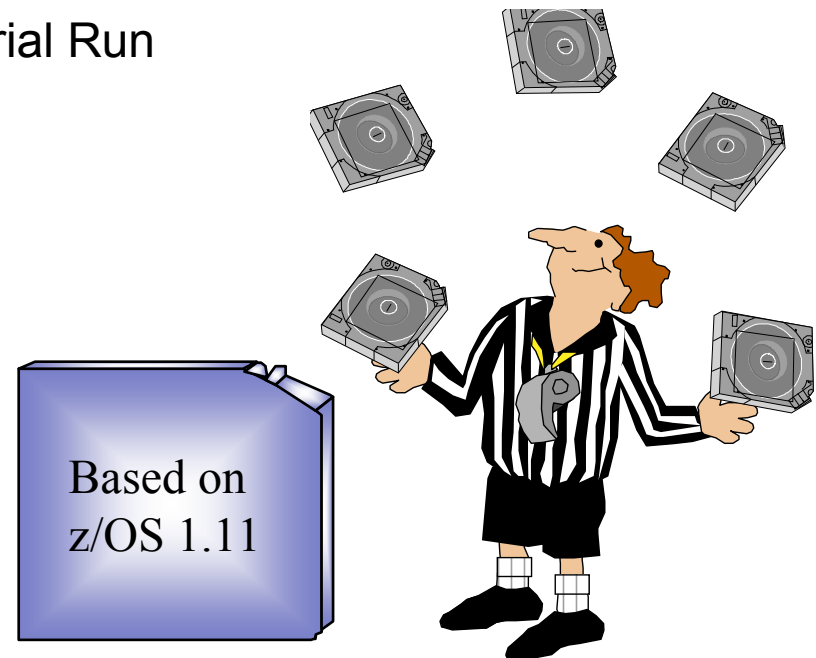
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Overview

- Retaining Data and Managing Life Cycles
- Policy Implementation
 - Inventory management
- Vital Record Specifications (VRS)
- Matching Policies to
 - Data Sets
 - Volumes
 - SMS Management Class
 - Management Value
- Retention Management
 - Retention Types
 - Multiple Retention Types
 - Retention Subchains
- Location Management / Vaulting
 - Stacked volume support
- Parmlib options
 - Volume Set Management
- Policy Management
 - Trial Run



Retaining Data

- **Common Reasons for Retention on Tape**
 - Archiving
 - Compliance
 - Business Recovery and Continuity (BRC)
 - Versioning
- **Retention**
 - Enforceable or Flexible?
- **Lifecycle Management**
 - Data Sets
 - From Creation through Expiration
 - Volumes
 - From Manufacture through Deployment and Retirement/Replacement

Defining Retention Policies

- **Select A Strategy**
 - Enforce policies using VRS
 - User Specifies Policy in JCL
 - Support a Combination based on need
- **Enforce Policies using**
 - RELEASEOPTIONS
 - MASTEROVERWRITE
 - RACF FACILITY CLASS
- **Support User Policy Selection Using**
 - JCL RETPD/EXPDT/ACCODE
 - For example, EXPDT=99000
 - Parmlib MAXRETPD Can Control RETPD Values
 - SMSACS/EDGUX100 may be Required for EXPDT/ACCODE special values
- **Decide The Service Levels Required**
 - Each Service Level is implemented using a VRS chain
- **Set Expectations**
 - How Much new Data is Policy Retained
 - How Much Policy Retained Data Expires

Setting Service Levels

- **Reason For Retention**
 - Archiving, Compliance, BRC, ...
- **Enforceable or Flexible**
 - Read only
 - Allow/Prevent Ignore/EXPDT=98000
- **Type of Retention**
 - Time, Copies, Catalog, ...
- **Where to be Stored**
 - Local, vault, VTS, ...
- **Identifying Data**
 - Application, Data Set Name Mask, ...
- **Define the VRS Chain**
- **Determine Parmlib Options**

Set Retention Expectations

- **Avoiding Loss of Data**

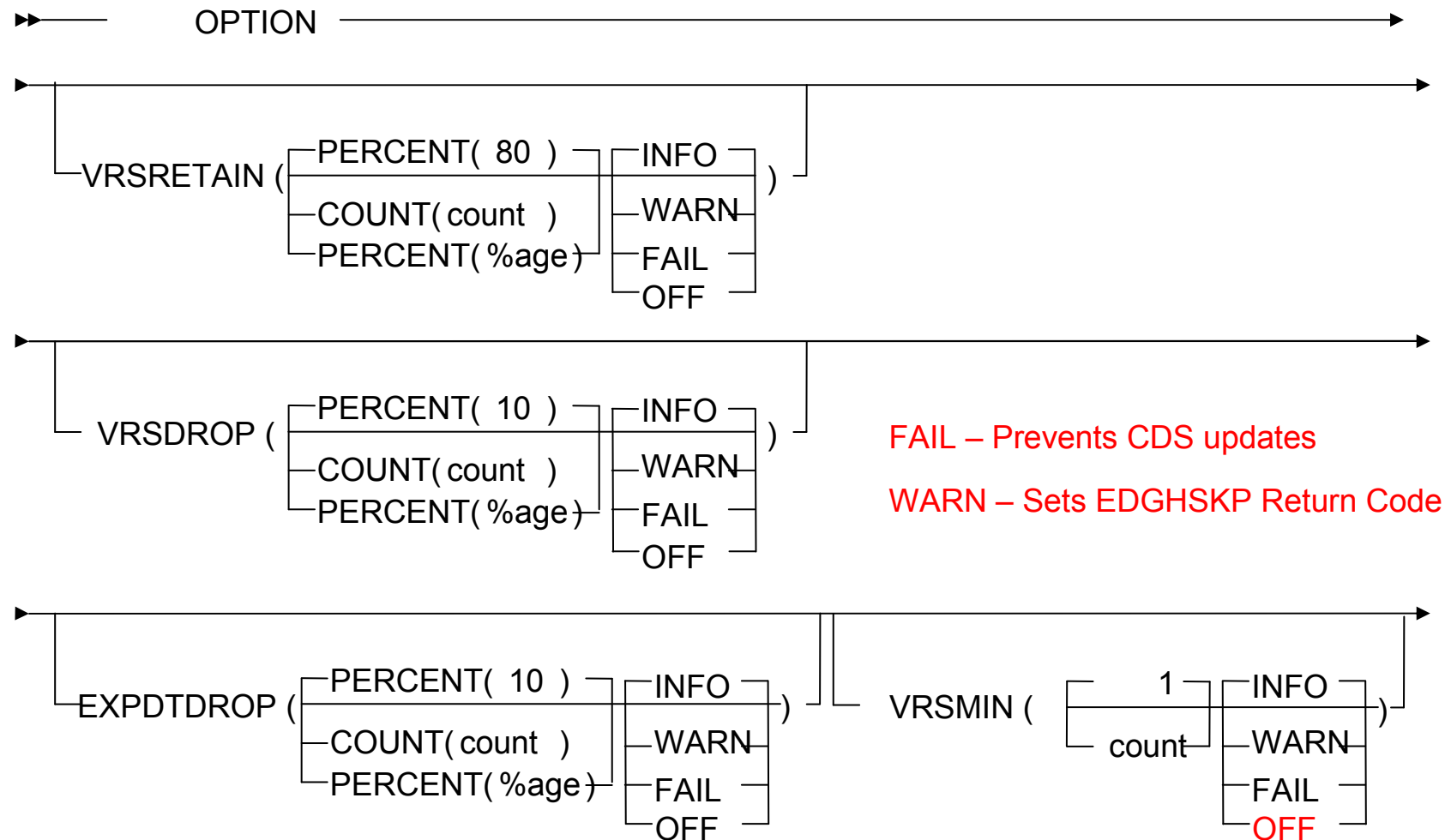
- Pre-empt system errors, user errors
- Define expected retention as an overall control
- Exploit automated way, to detect when too many volumes are released or expired

- **New parmlib options are added with z/OS V1.10**

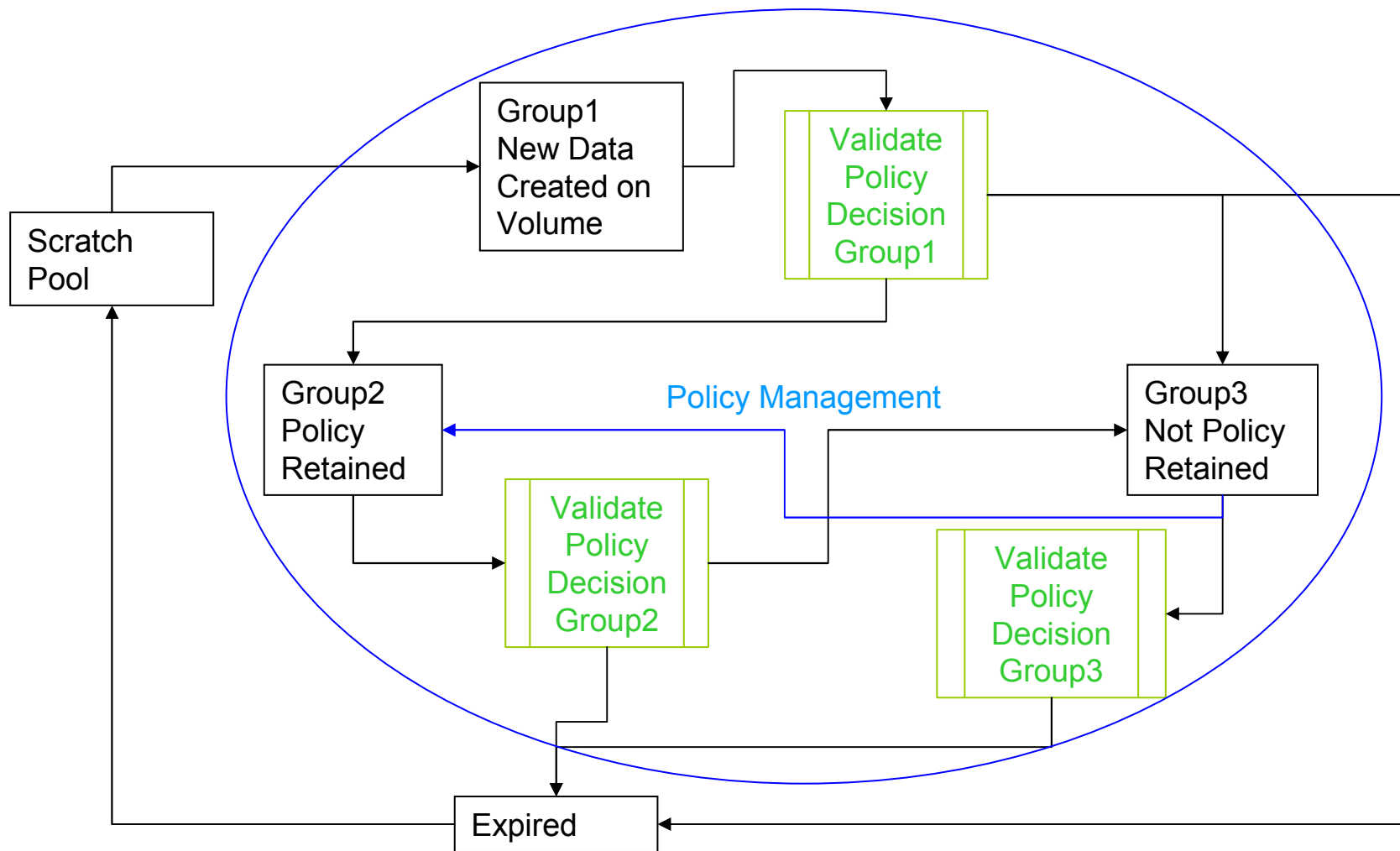
- Use VRSDROP to control volumes dropped from VRS retention
- Use VRSRETAIN to control newly assigned volumes that must be retained by VRS
- Use EXPDTDROP to control volumes dropped from EXPDT retention

- **Reporting of exceptions added via APAR OA30661**

Set Retention Expectations . . .



Life Cycle for Removable Media Data



Set Retention Expectations . . .

■ MESSAGE file

```
.....
EDG2420I PHYSICAL VOLUMES READ           =          150  75% ←
EDG2420I LOGICAL VOLUMES READ             =           30  15% ←
EDG2420I STACKED VOLUMES READ              =           20  10% ←
EDG2420I TOTAL VOLUMES READ                =          200 100% ←
.....
EDG2242I INITIAL NUMBER OF VRS RETAINED VOLUMES =          20  40% ←
EDG2244I NUMBER OF VRS RETAINED VOLUMES TO BE DROPPED =           2  10% ←
EDG2243I INITIAL NUMBER OF NEWLY ASSIGNED VOLUMES =          15  30% ←
EDG2245I NUMBER OF NEWLY ASSIGNED VOLUMES TO BE RETAINED =           5  33% ←
EDG2427I INITIAL NUMBER OF EXPDT RETAINED VOLUMES =          10 100% ←
EDG2428I NUMBER OF EXPDT RETAINED VOLUMES TO BE DROPPED =           2  20% ←
.....
EDG2421I PHYSICAL VOLUMES UPDATED          =           5  33% ←
EDG2421I LOGICAL VOLUMES UPDATED           =           1  33% ←
EDG2421I STACKED VOLUMES UPDATED           =           1  50% ←
EDG2421I TOTAL VOLUMES UPDATED             =           7  35% ←
```

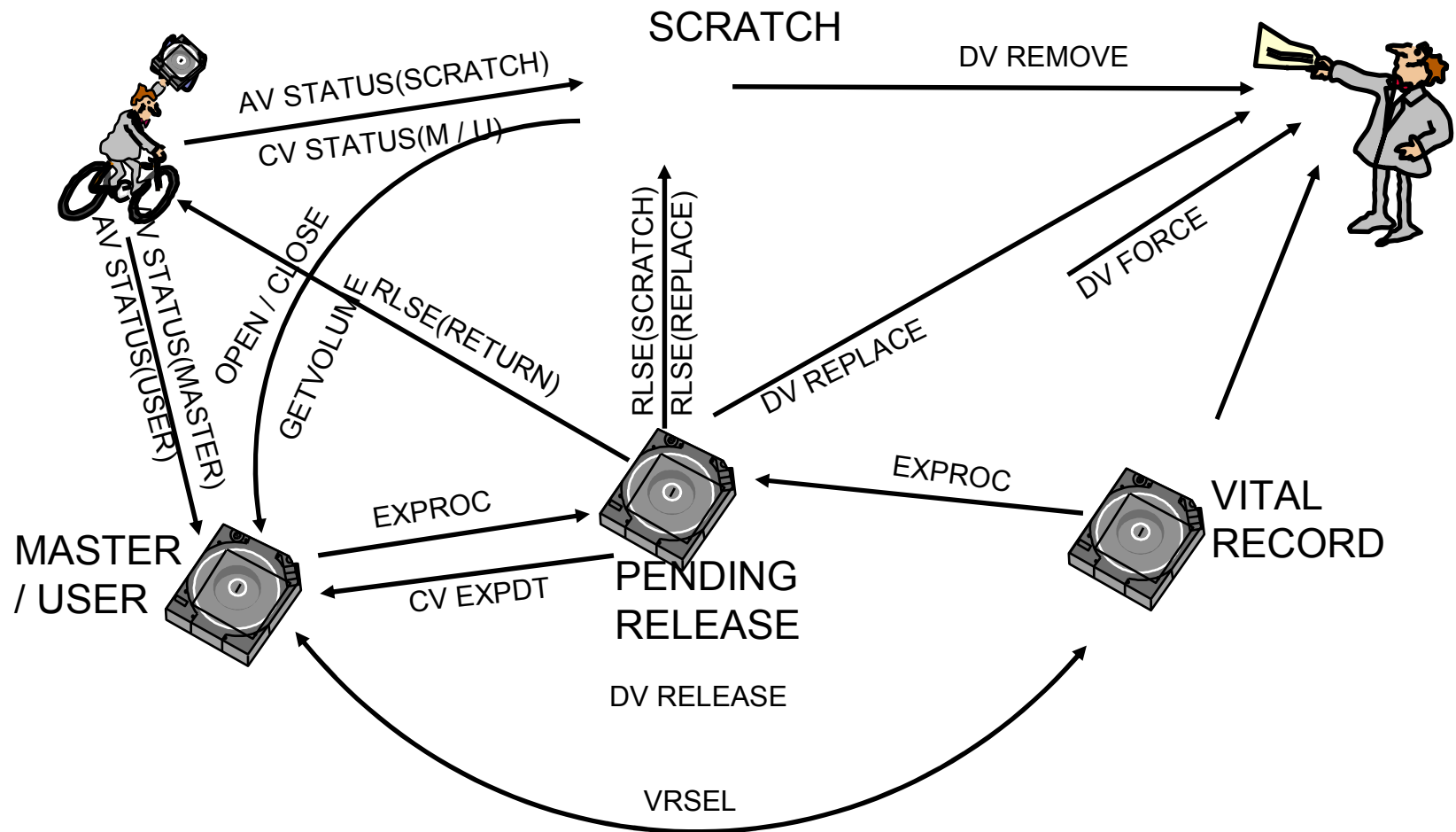
Archiving and Compliance

■ Special Needs for Enforcement

- Long term retention, prevention of Expiration
 - overridden temporarily for audit, investigations and legal reasons
 - Volume Level Volume VRS
 - Data Set Level MV/MC VRS
 - Pool Level EXPROC by Volume subset
- Prevent manual Release of Volumes
 - Limit Authorization via FACILITY CLASS
- Low Reference, Read only
 - OPTION/VLPOOL MASTEROVERWRITE(ADD) or WORM
- No Use Outside DFSMSrmm Control
 - Prevent 'Ignore' of Volume
 - Do not define OPENRULE with action(IGNORE)
 - EXPDT=98000
 - > Do not use EDGUX100
 - > Limit Authorization via FACILITY CLASS

R12 Adds HOLD
attribute for Volumes

Life Of A Volume



Media Life Cycle - Replacement

Prior to V1R10:

- Hard coded limit of '1' for permanent tape i/o errors Drives REPLACE release action
- USERMOD is available to set the limit to another value or disable support
- Non-Scratch Volumes are checked during Inventory Management and the REPLACE release action is set

V1R10 :

- New **Volume Replacement Policies** are added to parmlib to enable media end-of-life to be managed based on
 - permanent and temporary read and write errors detected by the tape hardware
 - volume usage
 - age
- RMM exploits tape drive **SARS MIM alerts** to identify active volumes which need replacement
- Non-Logical Volumes are checked during Inventory Management and the REPLACE release action is set
- Volumes with REPLACE release action are READ ONLY

Benefit:

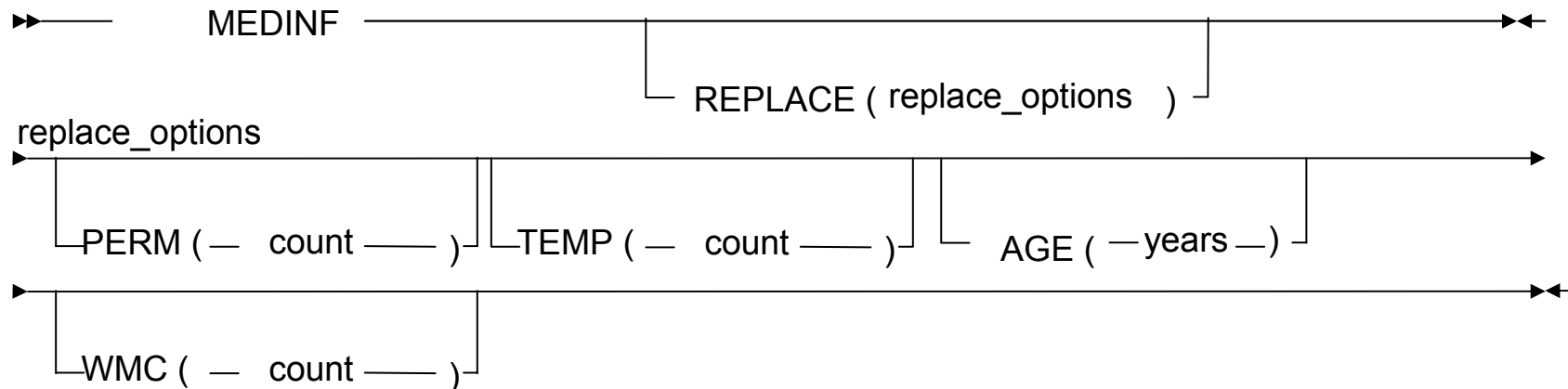
- Volume Replacement can be **easily customized**
- **Automated end-of-life management** of **active** and **inactive**, scratch and private volumes
- IBM Tivoli Tape Optimizer on z/OS V2.1 supports copying of non-expired volumes with the REPLACE release action

Media Life Cycle – Replacement . . .

■ Policies Can be Defined At:

- Default
 - REPLACE(PERM(1))
- Global Level
 - For example MEDINF NAME(IBM) REPLACE(PERM(0)) disables management
- Media Type
 - For example MEDINF NAME(ANO) MEDIATYPE(5,SQUARE) REPLACE(AGE(10))
- Specific Media Type and Recording Format
 - For example MEDINF NAME(ANO) MEDIATYPE(5,SQUARE) RECORDINGFORMAT(1,STD) REPLACE(WMC(999))

Go from
most specific
to less
specific



Media Life Cycle – Replacement . . .

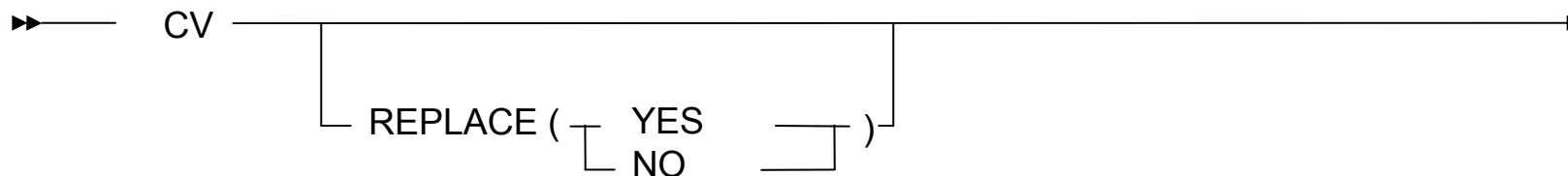
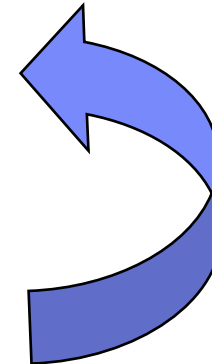
LC output of examples: Different types of Volume Replacement Policies

Media Information :									
	Name	Media type	Recording format	Capacity (MB)	Replace Policy				
					Perm	Temp	WMC	Age	
1. →	IBM	3 HPCT	0 *	0	1	5	5000	10	
2. →	NONIBM	1 STANDARD	3 36BTRACK	400	1	4	10000	15	
3. →	NONIBM	0 *	0 *	0	2	8	0	20	

- 1. Media type level policy
- 2. Specific policy
- 3. Global policy

Media Life Cycle – Replacement . . .

- **Inventory Management for REPLACE**
 - Physical, Stacked, WORM; Master, User, or Scratch
 - not pending release, release action SCRATCH or WORM volumes
 - Release Action set to REPLACE
- **SARS MIM for Active Volumes**
 - Release Action set to REPLACE
- **Follow up the REPLACE Action/Release Action**
 - Volume Expired and Pending Release
 - Replace the volume and confirm replacement
 - Cancel the REPLACE action
 - Delete the Volume
 - Volume Contains active Data
 - Copy Data to a Newer Volume
 - Release the Volume



Media Life Cycle – Replacement . . .

■ Exploitation of SARS MIM alerts

- **SARS** : Statistical Analysis & Recording System used by IBM tape drives for tracking media statistics to determine /predict media failure
- SARS Media Information Messages (**MIM**) alerts occur at the time a volume is used
 - An eventual action message issued to the console
 - WTO message **ID IEA486E** is intercepted by DFSMSrmm
 - Error reference code (**refc**) **60-62 or 64** is used to set volume release action from SCRATCH to **REPLACE**

IEA486E *cuu*, TVOL, {ACUTE | SERIOUS | MODERATE | SERVICE} ALERT,
VOLUME=*valid*, S=*s*, MC=*mc*, E=*e*, REF=*refc-med-fi*, REPEATED

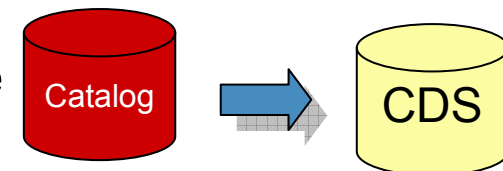
Policy Implementation

■ Inventory Management

- Vital Record Selection (PARM=VRSEL)
 - Dynamically applies Policies to Data Sets and Volumes and manages Retention and Movement



- Catalog Synchronization (PARM=CATSYNCH)
 - Sets Catalog Status for all Data Sets recorded in the CDS
 - VERIFY - reports on Differences
 - Runs once to Implement
 - Keeps in synch through catalog interface

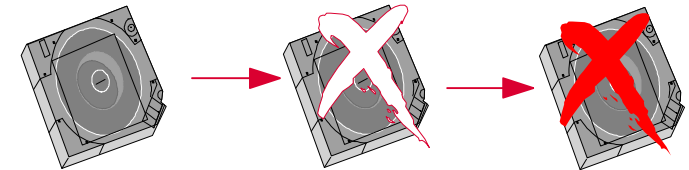


Policy Implementation . . .

■ Inventory Management . . .

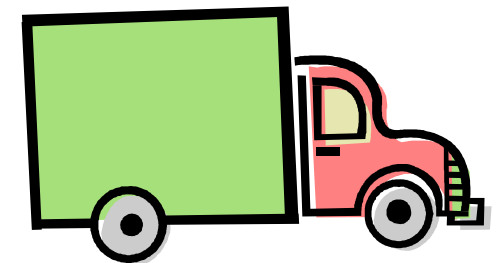
— Expiration Processing (PARM=EXPROC)

- Confirms Global Actions
- Implements Volume REPLACEment Policy
- Releases Volumes (Set Pending Release)
- Returns Volumes to Scratch
- Updates related information
 - Catalog
 - RACF
 - TCDB (SMS Tape Library)
- Optionally select volume subsets by volser and location



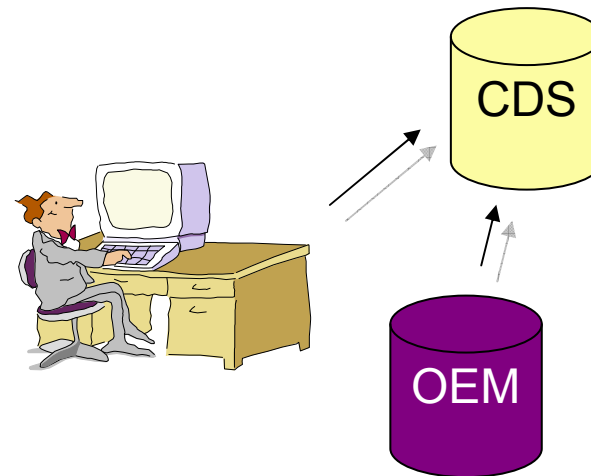
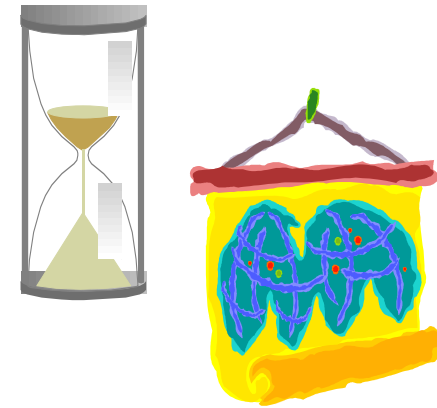
— Storage Location Management (PARM=DSTORE)

- Initiates/Confirms Moves
 - Optionally by Location
- Assigns and releases Bins
 - Optimized Bin Assignment to minimize slots required
 - Parmlib - REUSEBIN option



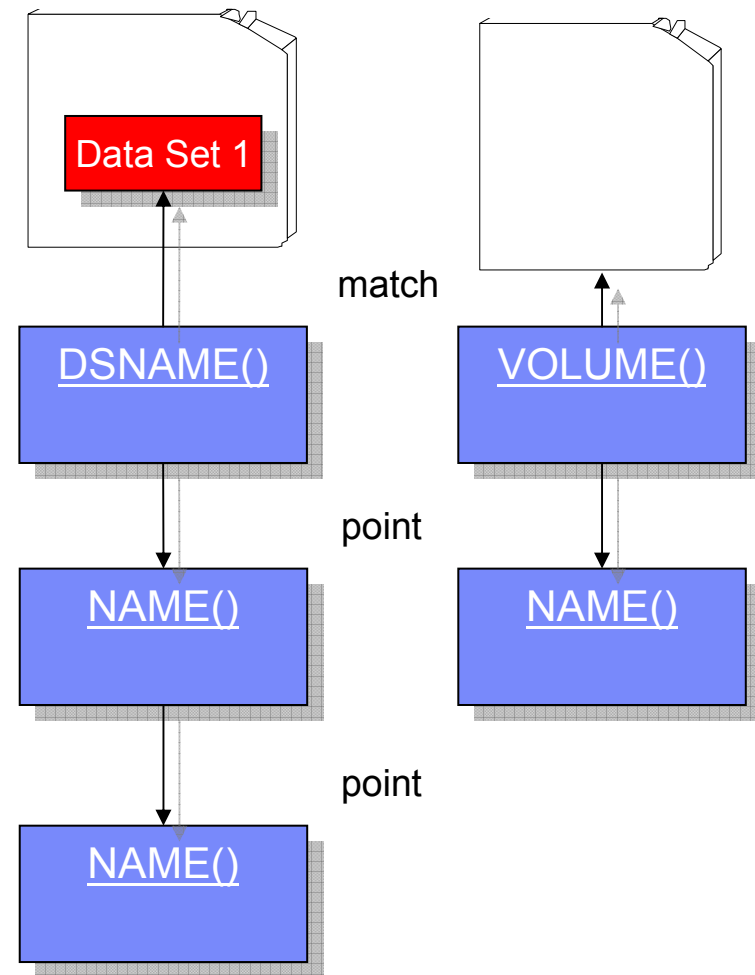
Vital Record Specification (VRS)

- **VRS's contain Policies to ...**
 - Control retention of data sets and volumes
 - Determine volume's required location
 - Set release options
- **VRS's are kept in the Control Data Set**
- **VRS's are created / maintained with**
 - Commands
 - ISPF Panels
 - Conversion tools



VRS Types

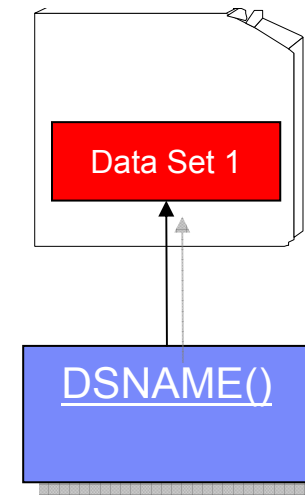
- **Data Set VRS**
 - Assigned to Data Sets
- **Volume VRS**
 - Assigned to Volumes
- **Name VRS**
 - Pointed by other VRS's
 - Retention Name VRS
 - Location Name VRS



Matching to a Data set VRS

■ Data Set Name

- Fully qualified data set name
- Data set name mask
- Single qualifier data set name mask
 - Management Class
 - Management Value
 - Reserved names
 - “OPEN”, “DELETED”, “ABEND”



■ Jobname

- Specific job name
- Job name mask
 - Reserved names
 - “OPEN”, “DELETED”, “ABEND”

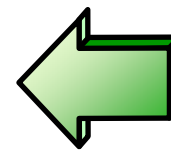
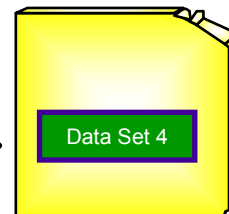
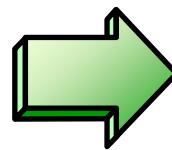
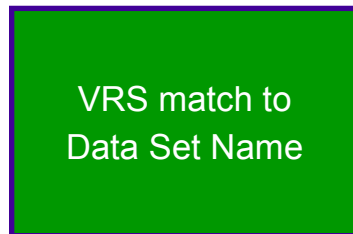
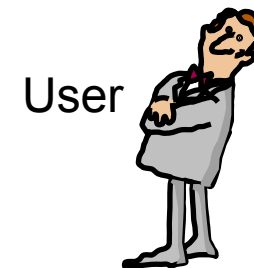
```
RMM ADDVRS DSNAME('PRITCHAR.BACKUP.DATA') DAYS COUNT(5)
RMM ADDVRS DSNAME('**') JOBNAME(LAURAN)
RMM ADDVRS DSNAME('**') JOBNAME(ABEND) DAYS COUNT(1)
```

■ GDG / NOGDG

Matching Order of Precedence

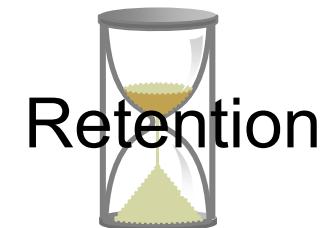
Primary VRS	Secondary VRS
OPEN VRS & Data Set is open	
DELETED VRS & DISP=DELETE	
ABEND VRS & Job abended	
Dsname VRS	Management Class VRS
Dsname VRS	Management Value VRS
Dsname VRS	
Management Class VRS	
Management Value VRS	
Default VRS – DSN('**')	

Option - Two Policies manage one data set



Primary VRS

Secondary VRS



Management Value

- **Management Value gets assigned to a Data Set by**
 - DFSMSRmm User Exit EDGUX100
 - Special Values in JCL parameter EXPDT=, ACCODE=
 - TSO RMM CHANGEDATASET command
 - Conversion to DFSMSRmm

Recommendation:

Use SMS Management Class to replace VRS Management Value

Management Class

- **Management Class gets assigned to a Data Set by**
 - SMS ACS (Automatic Class Selection)
 - JCL Parameter
 - TSO RMM CHANGEDATASET command
 - Conversion to DFSMSrmm

Recommendation:

Use SMS ACS FILTLISTs to replace RMM VRS Dsname Masks

Matching to a Volume VRS

- **By Specific Volume Serial Number**

- Retention is always DAYS since Creation of the first data set on the volume (assigned from scratch date)
- Retain the volume 10 days



VOLUME(ABC001)

COUNT(10)

- **By Generic Volume Serial Number**

- Retention is always CYCLES
- Retain the „newest“ 5 volumes from those starting with a Volser of ABC



VOLUME(ABC*)

COUNT(5)

Retention Types

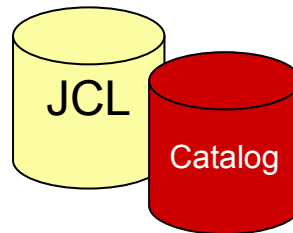
- Specific Date
- Days since Creation
- Days since Last Reference
- Extra Days



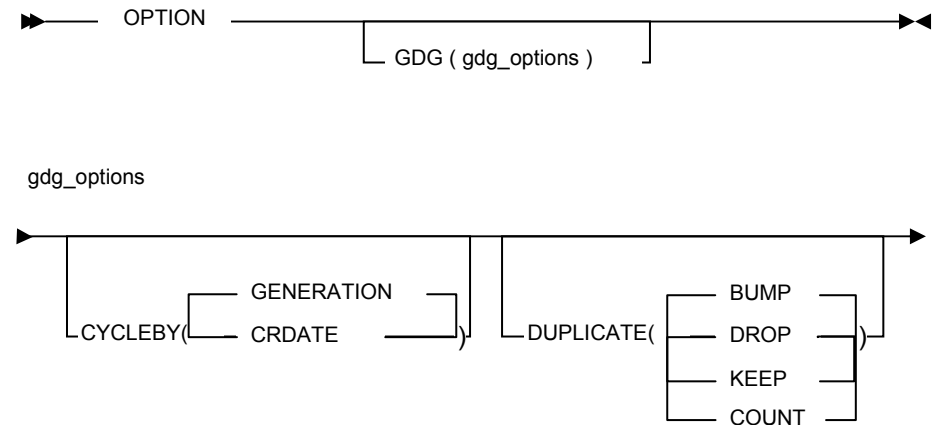
- Cycles
- By Days Cycles



- Until Expired
- While Cataloged

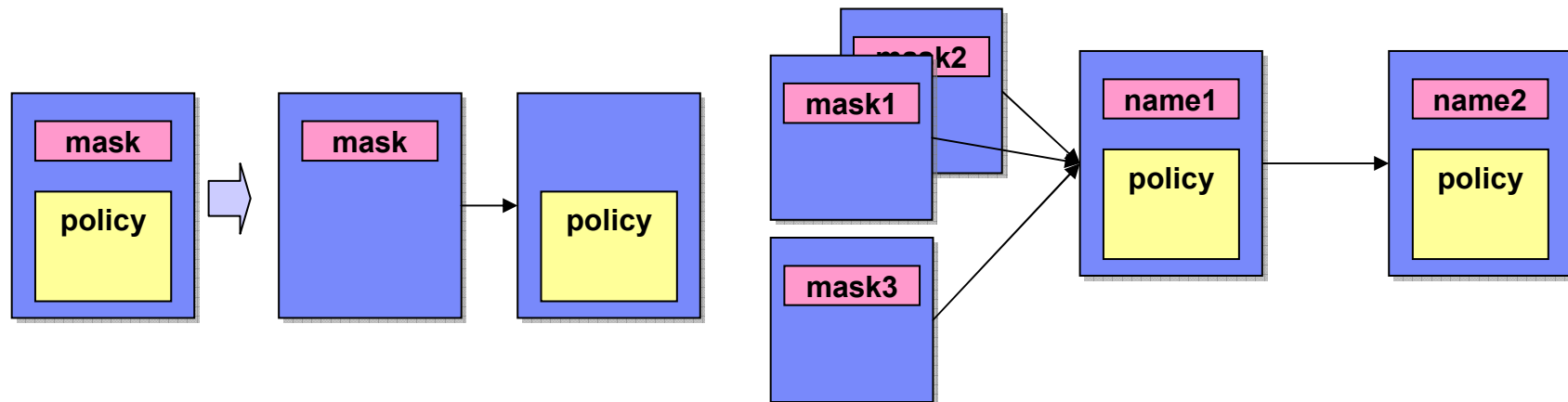


- Forever
- Don't retain



Reusable DSNNAME Policies

- **Separation** of the data set name-job name **mask** from the **policy** (retention/movement details)
- **Avoiding duplication** in VRS definitions
- When adding a new data set to VRS retention, the process is **simple** because you:
 - Select an existing **service level (policy)** represented by a VRS chain that starts with a retention NAME VRS
 - Define the data set name VRS, specifying DSNNAME, JOBNAME, and GDG/NOGDG as required, with **COUNT(0)** and **NEXTVRS(policy_name)**

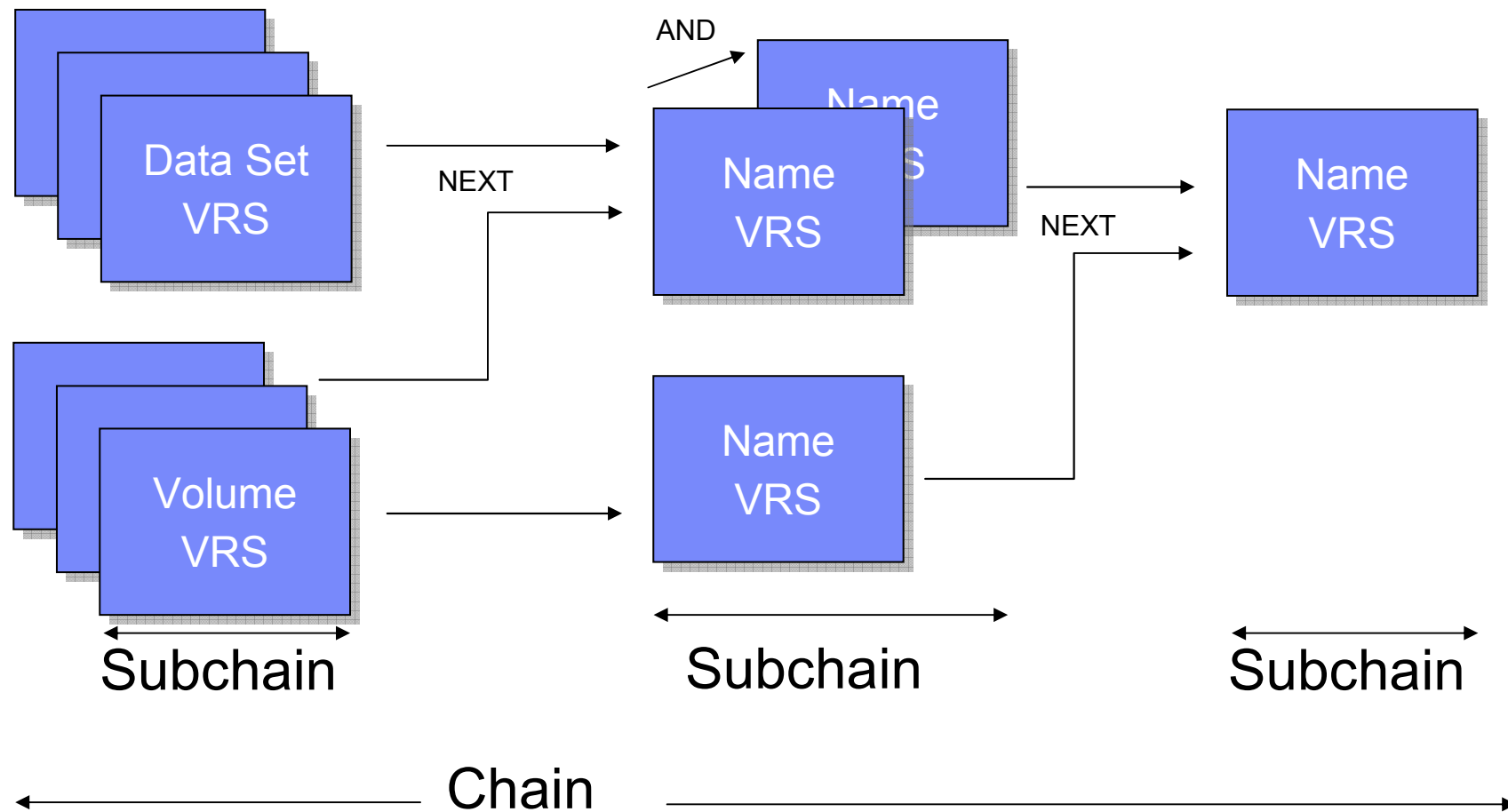


Multiple Retention Types

- **All retention criteria must be met to retain the data set**
 - Retain the data set while it is cataloged ...
 - until 100 days since creation

```
DSN('MC1')  
WHILECATALOG  
  
DAYS  
COUNT(100)
```

Chaining Retention and Movement Policies



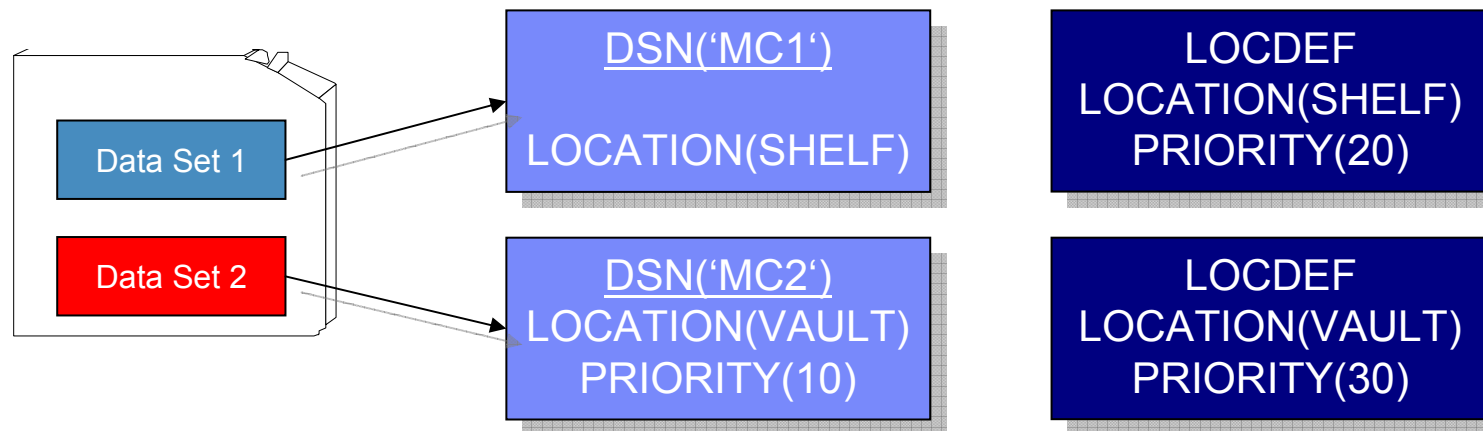
Location

- **Use Location parameter to specify “where” to retain the volume**
 - LOCATION(HOME)
 - Retain in the volume’s home location
 - LOCATION(CURRENT)
 - Retain in the current location
 - LOCATION(location_name)
 - Retain in the library or storage location specified
 - Built in storage locations: LOCAL, DISTANT, REMOTE
 - Installation defined storage locations defined with LOCDEF in the parmlib
- **VRSEL sets the Volume’s required location**
- **DSTORE sets the Volume’s destination and initiates the move**



Location Conflict

- **Solved by Location Priority defined in**
 - EDGRMMxx parmlib member – LOCDEF parameter
 - VRS (overrides LOCDEF)
- **Example: volume retained in VAULT**
 - Lowest number = highest priority



Retention Date

- **Data Set Retention Date**

- PERMANENT
- CYCL/cccc
- WHILECATLG
- date
- CATRETPD



- **Based upon VRS Subchain which currently retains the data set**

- **Volume Retention Date**

- Highest Retention Date of all Data Set Retention Dates

Release Options

- **Expiry Date Ignore**

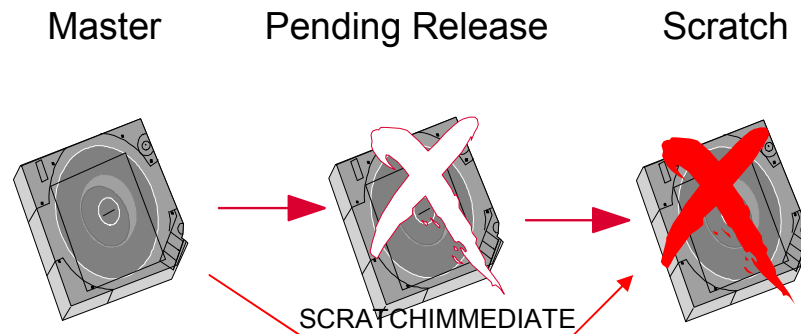
- Do not honor the volume's Expiry Date, once the volume is dropped by VRS

- **Scratch Immediate**

- Return the volume to Scratch immediately during release processing

```
DSN('MC1')  
  
RELEASE(  
EXPIRYDATEIGNORE)
```

```
DSN('MC1')  
  
RELEASE(  
SCRATCHIMMEDIATE)
```



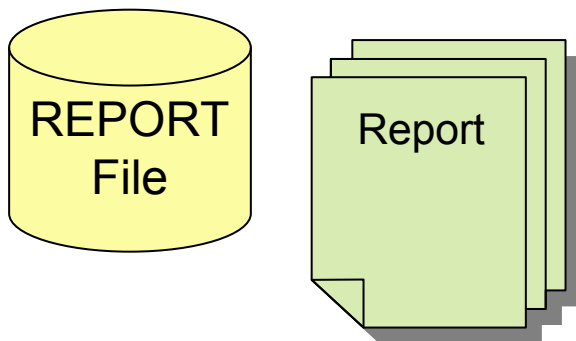
Release options for matching VRSs are applied regardless of whether the data set actually is ever VRS retained. The combination of COUNT(0) and release option changes ensures a Volume can be scratched the same day a Data Set on it is created. (Introduced with z/OS v1.8)

REPORT File

- **Contents**

- Formatted report of all retained data set and volumes
- Sorted by matching Primary VRS

- **Browse it or Print it**



- **Use to identify**

- groups of data sets
- cycles
- what is potentially moved

- **List of Unused VRSs**

- use it to identify and delete VRSs which are no longer required

ACTIVITY File

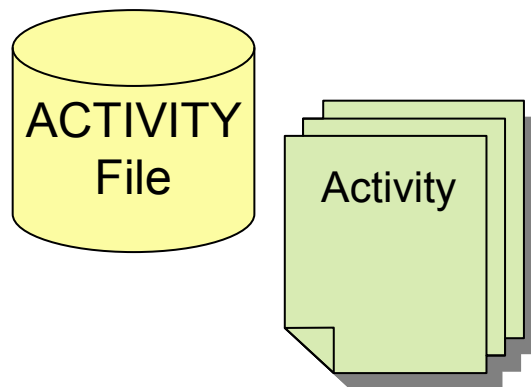
- **Contents**

- Header Record
 - Run Time Values
- Data set Record
 - Details of Changes

- **Browse it or Report on it**

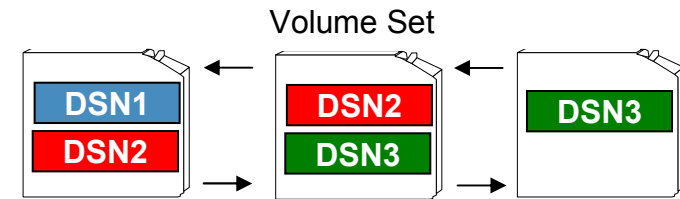
- **Sample Report**

- EDGJACTP
 - Vital status
 - Retention date
 - Matching VRS
 - VRS subchain
- Summaries
- Detailed reports



Recommendation:
Always write ACTIVITY File

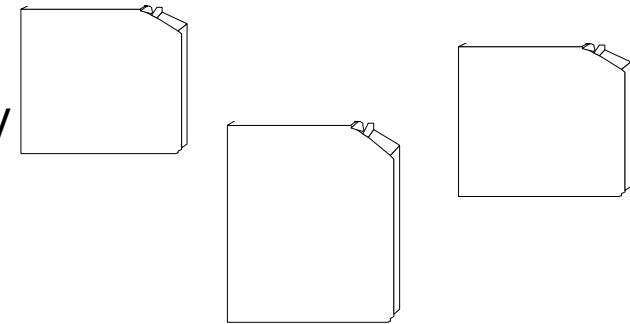
Parmlib Options



■ Volume Set Management

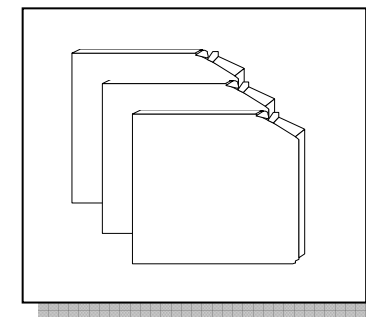
— RETAINBY(VOLUME)

- Volumes of a set are retained individually

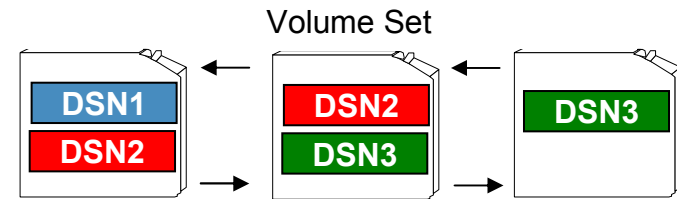


— RETAINBY(SET)

- If any volume in a set is retained by a VRS or Expiration date, all volumes in the set are retained.



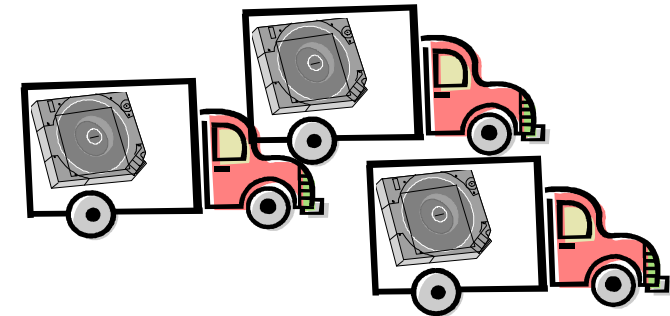
Parmlib Options



■ Volume Set Management

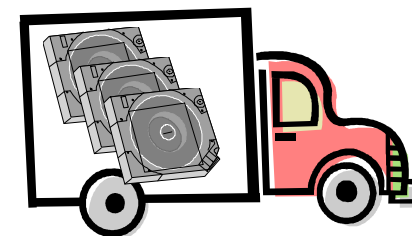
—MOVEBY(VOLUME)

- Volumes of a set are moved individually



—MOVEBY(SET)

- RMM Inventory Management sets the same required location for all volumes of a set which are retained by VRS
- The common location is determined by
 - VRS priority
 - Location priority



Parmlib Options

- **VRSEL(OLD | NEW)**

- NEW: should be used to exploit new functions
- OLD: different matching order, less functions
- The recommendation is to migrate from VRSEL(OLD) to VRSEL(NEW) before moving to z/OS V1R8

Removed in V1R11

- **VRSMIN(number, option)**

- Number: minimum number of policies (default 1)
- Option: Action if minimum not met
 - FAIL
 - WARN
 - INFO
 - OFF

- **VRSCCHANGE(option)**

- Option: first VRSEL run after change of a VRS
 - VERIFY
 - INFO - means: run as production

- **GDG(CYCLEBY()) DUPLICATE()**

- Controls for setting how GDGs are retained

Parmlib Options

- **CATRETPD(hours)**
 - Hours: catalog retention period (grace period)
- **RETPD(nnnn)**
 - Default retention period for data sets with no RETPD/EXPDT specified in JCL
- **MAXRETPD(NOLIMIT | nnnn)**
 - Controls maximum retention allowed in JCL
- **VRSRETAIN(number, option)**
 - Number: minimum number/percent of newly assigned volumes
 - Option: Action if minimum not met: FAIL, WARN, INFO, OFF
- **VRSDROP(number, option)**
 - Number: minimum number/percent of VRS retained volumes
 - Option: Action if minimum not met: FAIL, WARN, INFO, OFF
- **EXPDTDROP(number, option)**
 - Number: minimum number/percent of EXPDT retained volumes
 - Option: Action if minimum not met: FAIL, WARN, INFO, OFF

Policy Management Trial Run

- **Verify Policy changes**
- **Use to validate VRS adds, changes, deletes**
- **Protects production environment**
- **No CDS updates**
- **Can be used with future date**
- **Forecast retention changes**
- **Activity file required for VERIFY runs**

Recommendation:
Make use of Trial Runs

Example:

```
//STEP EXEC PGM=EDGHSKP,PARM='VRSEL,VERIFY,DATE=(+10)'
```

Summary of the tasks to perform

- **Agree Service Levels with Application and User Groups**
 - From these you can determine values for the following:
- **Set installation-wide retention periods in the parmlib member:**
 - RETPD, CATRETPD, MAXRETPD
- **Set other vital record processing options in the parmlib member:**
 - MOVEBY, RETAINBY, GDG, VRSJOBNAME, and VRSEL
- **Set Policy Control options in the parmlib member:**
 - VRSCCHANGE, VRSMIN, VRSRETAIN, VRSDROP, and EXPDTDROP
- **Set Media Replacement Rules via MEDINF**
 - Automate Replacement/Migration of Media
- **Define vital record specifications to retain and move data sets and volumes**
- **Define/Create data sets and volumes for DFSMSrmm to manage**
- **Optionally run inventory management in trial run mode**
- **Run DFSMSrmm inventory management vital record processing to process the policies you have defined**

z/OS DFSMSrmm V1R12

■ Policy Management related:

— Retention limit reporting

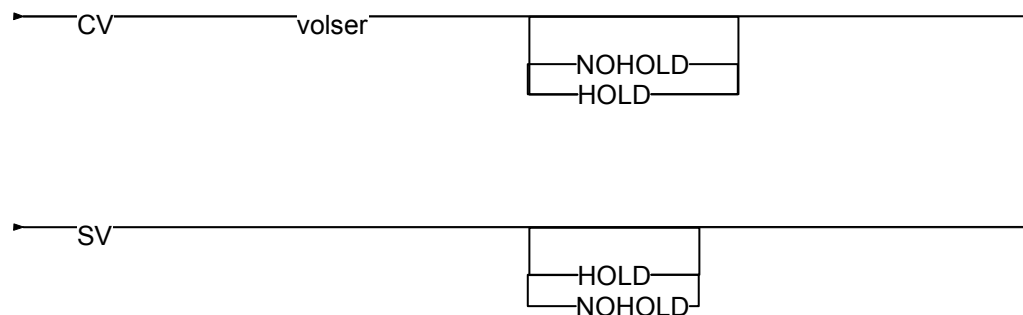
- EDGJACTP SAMPLIB JCL

(Roll-back to R10 via
APAR OA30881)

- New Reports: EXPDROP(S), VRSRETN(S)

- Existing Reports: VRS(S)

— New volume 'HOLD' attribute to prevent expiration/release



Statements of Future Plans and General Intention. Plans are Subject to Change

z/OS DFSMSrmm V1R12 . . .

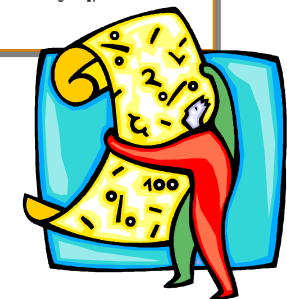
An Example of VRSRETAIN Report

Newly assigned volumes subject to VRSRETAIN										01/20/09	05:55:21	- 1 -	
Status: RETAINED													
D A T A S E T				D A T A S E T V R S			V O L U M E						
VOLSER	FSEQ	DSNAME	JOBNAME	RETAINED	DROP PRIM	REASON 2nd.	PRIMARY VRS	JOB MASK	VRS TYPE	VRS	RETAIN REASON	FILE COUNT	IN SET
VOL1	1	RMMUSER.DSN11		Y			RMMUSER.*		D		DATASET	3	N
VOL1	2	RMMUSER.DSN12		N	W		RMMUSER.*		D		IMPLICIT	3	N
VOL1	3	RMMUSER.DSN13		Y			RMMUSER.*		D		DATASET	3	N
VOL2	1	D046059.DSN21		N	W		D046059.*		D	VOL2	VOLUME	2	N
VOL2	2	D010155.DSN22								VOL2	VOLUME	2	N
VOL6	1	D046059.DSN61		N	D		D046059.*		D		SET	1	Y
VOL7	1	D077077.DSN71									IMPLICIT	2	Y
VOL7	2	D077077.DSN72		Y			D077077.DSN72		D		DATASET	2	Y
data sets in this status:				8									
Newly assigned volumes subject to VRSRETAIN										01/20/09	05:55:21	- 2 -	
Status: NOTRETAINED													
D A T A S E T				D A T A S E T V R S			V O L U M E						
VOLSER	FSEQ	DSNAME	JOBNAME	RETAINED	DROP PRIM	REASON 2nd.	PRIMARY VRS	JOB MASK	VRS TYPE	VRS	RETAIN REASON	FILE COUNT	IN SET
VOL3	1	RMMUSER.DSN31	STEINHA	N	W		RMMUSER.*		D			1	Y
VOL4	1	D010155.DSN41										2	Y
VOL4	2	RMMUSER.DSN42		N	W		RMMUSER.*		D			2	Y
VOL5	1	D010155.DSN51										3	N
VOL5	1	D010155.DSN52										3	N
VOL5	1	D010155.DSN53										3	N
data sets in this status:				6									

Summary of newly assigned volumes for VRSRETAIN

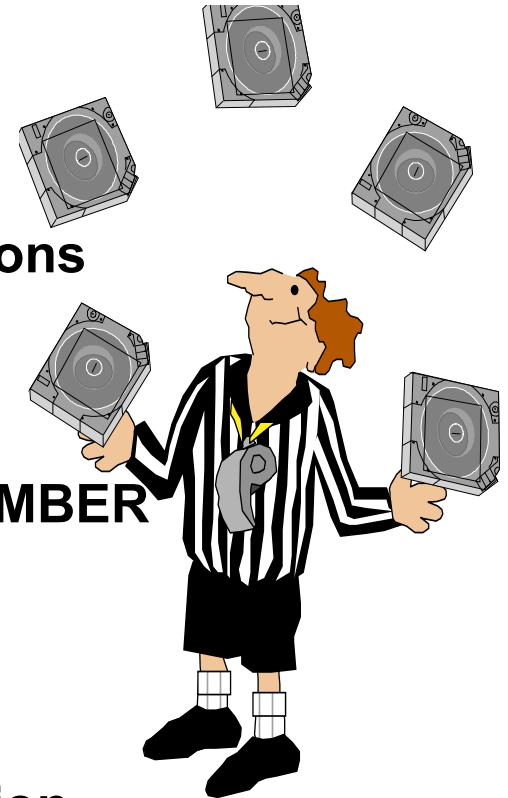
01/20/0905:55:21- 1 -

Status	VOLUME COUNT
RETAINED	4
NOTRETAINED	3



Recommendations

- **Exploit SMS Management Class or DSNAME mask VRS's**
- **Establish Service Levels for**
 - Retention and
 - Vaulting
- **Set Retention Expectations and Exception Actions**
- **Reduce number of VRS's**
- **Convert to VRSEL(NEW)**
- **Use Retention Subchains instead of STORENUMBER**
- **Implement CATSYNCH**
- **Set Volume Replacement Rules and Automate Replacement**
- **Consider Enforcement Needs Based on Retention Reasons**



Last but not Least

- **Everything should be made as simple as possible, but not simpler**

